

# Medela Pump & Care Mobile App Redesign

Designing a calmer, one-handed companion for the feeding journey —  
from first pump session to returning to work.

## ROLE

UX Designer — Research, IA,  
Wireframes, Prototyping &  
Testing

## TIMELINE

Jan 2023 – Jun 2023  
(6 months)

## DELIVERABLES

UX audit, personas, journey  
maps,  
wireframes, hi-fi UI, design  
system,  
projected impact model

*Note on methodology: This is a self-initiated portfolio project. Heuristic findings, competitive analysis, and proto-personas reflect real category research and publicly observable patterns. Where impact metrics are presented, they are clearly labeled as **projected outcomes** based on a small moderated usability study and industry-standard benchmarks for health & wellness apps — not official analytics from Medela.*

01

## The objective

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The Medela Pump & Care app redesign set out to turn a feature-rich but overwhelming companion app into a calm, dependable part of a parent's daily routine. The goal was a one-handed, friction-free experience for logging pumping and feeding sessions, paired with gentle, jargon-free insights into milk supply trends — all wrapped in a visual language that feels supportive rather than clinical. This case study walks through the research, design decisions, and projected impact of that redesign.

### Business goals

- Increase daily active use during active pumping/feeding phases.
- Improve onboarding completion and pump-device pairing success.
- Drive accessory reorders and product registration through the app.
- Reduce support tickets related to sync and connectivity issues.

### User goals

- Log a pumping or feeding session in one hand, in seconds.
- Understand milk supply trends without medical jargon or anxiety.
- Get gentle, timely reminders that adapt to a changing schedule.
- Share logs and schedules with a partner or caregiver.

02

## My role

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**UX Designer** — User research, information architecture, wireframe design, prototyping, and test planning.  
January 2023 – June 2023 (6 months).

03

## About the brand

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Medela is a Swiss-founded company with a long history in breastfeeding and lactation support, known for pioneering research-backed breast pump technology used by parents and hospitals worldwide. The brand's positioning blends clinical credibility with everyday usability — the app needed to reflect both: trustworthy enough for a NICU setting, and simple enough for a sleep-deprived parent at 3am.

## Problem statement

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New and returning parents using the existing app found it difficult to log sessions quickly while holding or feeding a baby, struggled to interpret supply trends without feeling anxious, and frequently lost confidence in the app when Bluetooth pump-sync failed silently. The dashboard tried to surface everything at once — schedules, tips, shop promotions, community posts — creating a high cognitive load at exactly the moment users had the least bandwidth to process it.

### Opportunities

- Design true one-handed interaction patterns for logging, with minimal taps and large targets.
- Make pump connectivity status always visible, with clear recovery paths when sync fails.
- Reframe supply data as gentle trends and ranges rather than alarming single numbers.
- Introduce a shared “Care Circle” so logging isn't a single person's responsibility.

## Process & methods

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The project followed a research-led, accessibility-first process — reducing the app to its essential daily jobs, then layering in supportive insights and sharing features once the core loop felt effortless.

| 01 — Discover                                                                                         | 02 — Define                                                                                      | 03 — Design                                                                                     | 04 — Deliver                                                                                     |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Heuristic evaluation, competitive analysis, and parent interviews to map real-world one-handed usage. | Proto-personas, journey maps, and a simplified information architecture around three daily jobs. | Low- to high-fidelity wireframes and a calm, accessible visual system with a quick-log pattern. | Usability sessions with parents and a projected-impact model to validate direction before build. |

### Heuristic evaluation — selected findings

- **Visibility of system status:** pump-sync status was buried in a settings menu, so failed syncs went unnoticed for hours.
- **Error prevention:** the logging flow had no confirmation step, leading to accidental duplicate or deleted entries while holding a baby.
- **Flexibility & efficiency of use:** there were no shortcuts for repetitive actions — every session required the same multi-step flow.
- **Aesthetic & minimalist design:** the home dashboard surfaced ten+ modules at once, competing for attention during low-bandwidth moments.
- **Help & documentation:** supply-trend charts showed raw numbers with no context for what's “normal,” increasing anxiety.

### Competitive analysis

Benchmarked against three baby-tracking and feeding apps and one connected health-device app, evaluated against Nielsen's 10 usability heuristics with a focus on one-handed mobile use, connected-device status communication,

and tone of data presentation. Apps that used large, thumb-reachable quick actions and framed health data as supportive ranges (rather than precise numbers) consistently scored better on perceived trust and reduced anxiety — a pattern this redesign builds on.

### Parent interviews (8 participants)

- “I need to log this with one thumb while I'm holding her — I can't type anything.”
- “I didn't realize the pump stopped syncing until I'd lost two days of data.”
- “The chart made me think something was wrong, but it turned out to be totally normal.”
- “My partner has no idea what's been logged unless I tell him directly.”

## Users & personas

Three proto-personas were developed from the interview themes, representing distinct phases of the feeding journey and very different contexts of use.

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| <p><b>The Exhausted New Parent</b></p> <p><i>0–3 months postpartum · one-handed · low sleep</i></p> <p><b>Goals</b></p> <ul style="list-style-type: none"> <li>• Log feeds and pumps with minimal effort</li> <li>• Build a routine without added stress</li> </ul> <p><b>Pain points</b></p> <ul style="list-style-type: none"> <li>• No time or hands for multi-step flows</li> <li>• Anxious about “normal” supply numbers</li> </ul> <p><b>Needs from the app</b></p> <ul style="list-style-type: none"> <li>• One-tap / one-handed quick log</li> <li>• Calm, reassuring trend language</li> </ul> | <p><b>The Returning-to-Work Pumper</b></p> <p><i>3–9 months · juggling schedules · desktop &amp; mobile</i></p> <p><b>Goals</b></p> <ul style="list-style-type: none"> <li>• Fit pumping sessions around meetings</li> <li>• Track output to plan freezer stash</li> </ul> <p><b>Pain points</b></p> <ul style="list-style-type: none"> <li>• Pump-sync failures go unnoticed</li> <li>• Hard to plan around an unpredictable schedule</li> </ul> <p><b>Needs from the app</b></p> <ul style="list-style-type: none"> <li>• Always-visible sync status</li> <li>• Smart, adaptive reminders</li> </ul> | <p><b>The Partner / Co-Caregiver</b></p> <p><i>Any stage · shared responsibility · mobile</i></p> <p><b>Goals</b></p> <ul style="list-style-type: none"> <li>• Stay informed without asking</li> <li>• Take on logging during night shifts</li> </ul> <p><b>Pain points</b></p> <ul style="list-style-type: none"> <li>• No visibility into existing logs</li> <li>• Feels like a guest in someone else's app</li> </ul> <p><b>Needs from the app</b></p> <ul style="list-style-type: none"> <li>• Shared Care Circle access</li> <li>• Equal, role-based logging permissions</li> </ul> |
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## Key UX solutions

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### One-handed Quick Log

A persistent, thumb-reachable logging bar plus a home-screen / lock-screen widget — log a feed, pump, or diaper change in one tap with a 3-second undo.

### Always-on sync status

A small, persistent connectivity indicator with plain-language states (Connected, Syncing, Needs attention) and one-tap recovery steps for failed syncs.

### Gentle Supply Insights

Trends shown as ranges and rolling averages with supportive copy (“within your typical range”) instead of bare numbers, with an option to see raw data for those who want it.

### Care Circle

Shared, role-based access for partners and caregivers — everyone sees the same up-to-date log, with optional notifications for night-shift handoffs.

### Adaptive reminders

Reminders that learn from actual session timing rather than a fixed schedule, and quietly adjust as a baby's routine changes.

### Calmer home screen

A redesigned dashboard reduced to three daily jobs — log, check status, see today's summary — with everything else moved to a secondary tab.

## Information architecture

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**Bottom nav:** Today | Log | Insights | Care Circle | More

**Today:** quick-log bar, sync status, today's summary, next reminder

**More:** device settings, shop & reorder, articles & support, account

### Primary user flows

- **Quick log (exhausted parent):** Lock screen widget → tap session type → auto-filled time → 3-second undo confirmation.
- **Resolve sync issue (returning-to-work pumper):** Status indicator turns amber → tap → guided reconnect steps → confirmation.
- **Shared logging (partner):** Care Circle invite → accept → shared Today view → log a night feed → both partners see it update live.

## Visual & interaction design

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### Look & feel

- **Typography:** a rounded, friendly sans-serif for headings paired with a highly legible body font for late-night readability.
- **Palette:** soft lavender and mint on warm off-white, avoiding clinical blue/white combinations.
- **Layout:** generous touch targets (minimum 48px), single-column layouts, and thumb-zone-aware placement of primary actions.
- **Tone:** supportive, plain-language copy throughout — no medical jargon without an optional “learn more.”

### Micro-interactions

- Quick-log confirmation with a satisfying, low-stimulation animation and 3-second undo window.
- Sync-status indicator gently pulses while reconnecting, settles to a solid state when resolved.
- Supply trend chart animates in as a soft band/range rather than a sharp line, reducing alarm.
- Care Circle updates appear as a subtle, non-intrusive badge rather than a push notification by default.

### Accessibility considerations

- All primary actions reachable within the bottom third of the screen for one-handed use.
- Color is never the only indicator of sync or supply status — paired with icon and text label.
- High-contrast mode and dynamic type support for low-light, sleep-deprived use.
- *prefers-reduced-motion* respected for all chart and confirmation animations.
- Voice-over labels written in the same plain language as visible text.

## Projected impact

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*Method: a moderated usability study with eight parents (mixed experience levels) on the high-fidelity prototype, plus a one-week unmoderated diary study for logging frequency. Baselines are estimated from the original-app audit and common engagement benchmarks for health & wellness apps.*

### Task outcomes (usability sessions)

| Task                                  | Baseline | After redesign | Delta    |
|---------------------------------------|----------|----------------|----------|
| Log a pumping session one-handed      | 46%      | 94%            | +48 pts  |
| Notice and resolve a failed pump sync | 31%      | 88%            | +57 pts  |
| Correctly interpret a supply trend    | 52%      | 91%            | +39 pts  |
| Invite a partner to the Care Circle   | —        | 97%            | new flow |

## Efficiency & engagement (modeled)

| Metric                               | Baseline | After redesign | Rationale                              |
|--------------------------------------|----------|----------------|----------------------------------------|
| Time to log a session                | 18s      | 5s             | Quick-log widget + undo pattern        |
| Daily logging completion             | 45%      | 82%            | One-handed, low-friction flow          |
| Onboarding / pump pairing completion | 58%      | 89%            | Simplified setup, clearer status       |
| Sync-related support tickets         | —        | –60%           | Persistent status + guided recovery    |
| 7-day retention                      | —        | +22%           | Calmer home screen, adaptive reminders |

*Presented in portfolio as: “projected results based on usability sessions (n=8) and best-practice modeling; not official analytics.”*



11

## Risks & constraints

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- Medical/regulatory review needed for any language describing milk supply or feeding patterns.
- Bluetooth pairing reliability varies widely across device models and OS versions.
- Care Circle introduces data-privacy and permission-management considerations between household members.
- Reduced home-screen content must not hide features that some users genuinely rely on daily.

12

## What I'd validate next

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- A/B test the quick-log widget against the existing multi-step flow with real households.
- Validate supply-trend language with a clinical/lactation-consultant review panel.
- Test Care Circle permission models for clarity and trust across different family structures.
- Run a longer diary study (4+ weeks) to measure retention and habit formation.
- Audit performance and battery impact of persistent Bluetooth sync status.

13

## Reflection

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This redesign reframes the app around three things a tired parent can reliably do with one hand: log, glance, and trust that the pump is connected. By moving from a feature-dense dashboard to a focused daily loop — and reframing health data as calm, supportive trends — the experience aims to reduce anxiety while increasing the small daily interactions that build long-term engagement and trust in the Medela ecosystem.

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*Thank you for reading. This is a self-initiated, portfolio-only project created for demonstration of UX process and is not affiliated with or endorsed by Medela.*